

Question	Marking Guidance	Mark	Comments
10.1	1. Hydrogen bonds (between DNA bases) break; 2. (Only) one DNA strand acts as a template; 3. (Free) RNA nucleotides align by complementary base pairing; 4. (In RNA) Uracil base pairs with adenine (on DNA) OR (In RNA) Uracil is used in place of thymine; 5. RNA polymerase joins (adjacent RNA) nucleotides; 6. (By) phosphodiester bonds (between adjacent nucleotides); 7. Pre-mRNA is spliced (to form mRNA) OR Introns are removed (to form mRNA);	5 max	Ignore DNA helicase. 1. Reject hydrolysing hydrogen bonds. 3. For 'align by complementary base pairing', accept 'align to complementary bases' or 'align by base pairing'. 4. Do not credit use of letters alone for bases. 5. Reject suggestions that RNA polymerase forms hydrogen bonds or joins complementary bases.
10.2	1. (mRNA attaches) to ribosomes OR (mRNA attaches) to rough endoplasmic reticulum; 2. (tRNA) anticodons (bind to) complementary (mRNA) codons; 3. tRNA brings a specific amino acid; 4. Amino acids join by peptide bonds; 5. (Amino acids join together) with the use of ATP; 6. tRNA released (after amino acid joined to polypeptide); 7. The ribosome moves along the mRNA to form the polypeptide;	6 max	

10.3	(Definition of gene mutation) 1. Change in the base/nucleotide (sequence of chromosomes/DNA); 2. Results in the formation of new <u>allele</u>; (Has no effect because) 3. Genetic code is degenerate (so amino acid sequence may not change); OR Mutation is in an intron (so amino acid sequence may not change); 4. Does change amino acid but no effect on tertiary structure; 5. (New allele) is recessive so does not influence phenotype; (Has positive effect because) 6. Results in change in polypeptide that positively changes the properties (of the protein) OR Results in change in polypeptide that positively changes a named protein; 7. May result in increased reproductive success OR May result in increased survival (chances);	4 max	For 4 marks at least one mark must be scored in each section of the answer. 1. Accept named mutation for 'change'. 3. Accept description of 'degenerate', eg some amino acids have more than one triplet/codon. 6. For 'polypeptide' accept 'amino acid sequence' or 'protein'.
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Appendix 1

